**BDA LAB 3**
**16.02.2026**
**(1BM23CS310)**

## LIBRARY SYSTEM

```
cqlsh> CREATE KEYSPACE Library
   ... WITH replication = {
   ...     'class': 'SimpleStrategy',
   ...     'replication_factor': 1
   ... };
cqlsh> USE Library;
cqlsh:library> CREATE TABLE Library_Info (
           ...     Stud_Id int,
           ...     Book_Id int,
           ...     Stud_Name text,
           ...     Book_Name text,
           ...     Date_of_issue date,
           ...     PRIMARY KEY (Stud_Id, Book_Id)
           ... );
cqlsh:library> CREATE TABLE Library_Counter (
           ...     Stud_Id int,
           ...     Book_Name text,
           ...     Counter_value counter,
           ...     PRIMARY KEY (Stud_Id, Book_Name)
           ... );
cqlsh:library> BEGIN BATCH
           ...
           ... INSERT INTO Library_Info (Stud_Id, Book_Id, Stud_Name, Book_Name, Date_of_issue)
           ... VALUES (112, 101, 'Rahul', 'BDA', '2026-02-10');
           ...
           ... INSERT INTO Library_Info (Stud_Id, Book_Id, Stud_Name, Book_Name, Date_of_issue)
           ... VALUES (113, 102, 'Amit', 'ML', '2026-02-12');
           ...
           ... APPLY BATCH;
cqlsh:library> UPDATE Library_Counter
           ... SET Counter_value = Counter_value + 1
           ... WHERE Stud_Id = 112 AND Book_Name = 'BDA';
cqlsh:library>
cqlsh:library> UPDATE Library_Counter
           ... SET Counter_value = Counter_value + 1
           ... WHERE Stud_Id = 112 AND Book_Name = 'BDA';
```

```
cqlsh:library> DESCRIBE TABLE Library_Info;

CREATE TABLE library.library_info (
    stud_id int,
    book_id int,
    book_name text,
    date_of_issue date,
    stud_name text,
    PRIMARY KEY (stud_id, book_id)
) WITH CLUSTERING ORDER BY (book_id ASC)
    AND additional_write_policy = '99p'
    AND bloom_filter_fp_chance = 0.01
    AND caching = {'keys': 'ALL', 'rows_per_partition': 'NONE'}
    AND cdc = false
    AND comment = ''
    AND compaction = {'class': 'org.apache.cassandra.db.compaction.SizeTieredCompactionStrategy', 'max_threshold': '32', 'min_threshold': '4'}
    AND compression = {'chunk_length_in_kb': '16', 'class': 'org.apache.cassandra.io.compress.LZ4Compressor'}
    AND memtable = 'default'
    AND crc_check_chance = 1.0
    AND default_time_to_live = 0
    AND extensions = {}
    AND gc_grace_seconds = 864000
    AND max_index_interval = 2048
    AND memtable_flush_period_in_ms = 0
    AND min_index_interval = 128
    AND read_repair = 'BLOCKING'
    AND speculative_retry = '99p';
cqlsh:library> SELECT * FROM Library_Info;

 stud_id | book_id | book_name | date_of_issue | stud_name
---------+---------+-----------+---------------+-----------
     113 |     102 |        ML |    2026-02-12 |      Amit
     112 |     101 |       BDA |    2026-02-10 |     Rahul

(2 rows)
cqlsh:library> UPDATE Library_Counter
           ... SET Counter_value = Counter_value + 1
           ... WHERE Stud_Id = 112 AND Book_Name = 'BDA';
cqlsh:library> SELECT Counter_value
           ... FROM Library_Counter
           ... WHERE Stud_Id = 112 AND Book_Name = 'BDA';

 counter_value
---------------
             3
```

```
cqlsh:library> COPY Library_Info TO 'library_info.csv' WITH HEADER = TRUE;
Using 16 child processes

Starting copy of library.library_info with columns [stud_id, book_id, book_name, date_of_issue, stud_name].
Processed: 2 rows; Rate:      22 rows/s; Avg. rate:      22 rows/s
2 rows exported to 1 files in 0.107 seconds.
```